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Cloning the FlightGear Space Shuttle model to start work on an Orbitron fusion-powered spaceplane

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Open mind

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In a **previous article** I talked about making a fusion-power SSTO spaceplane. To recap, what I want is a

- Single-stage to orbit spaceplane.
- Energy for thrust generated by a 10 x 10 x 10 cube of hypothetical modular 1mW **Avalanche Energy Orbitron** fusion reactors running Proton-Boron-11 fuel generating 1GW of electricity.
- Reaction mass for thrust is air when in the atmosphere and then water when in vacuum.
- The engine is a 4-stage SABRE model.
- *Maybe* make the cockpit more like an embedded **SpaceX** Crew Dragon capsule or a **Lockheed Martin** Orion capsule and less like **NASA - National Aeronautics and Space Administration** Space

Shuttle. But not necessarily. The Space Shuttle had a toilet. The Crew Dragon has a toilet. For some reason this became a **hard problem** on the Orion.

- Pretty much the space occupied by the entire Space Shuttle payload bay would be occupied by air ducts, internals of SABRE engines, the array of Orbitrons, the large water supply and small p-B11 supply. So it's not a cargo-carrying spacecraft. It's main goal is SSTO for the crew and cargo stored in the crew capsule.

The 4 stages of the engine are

1. **First Gear (Runway):** The engines operate in "bypass mode." The 1 GW of electricity is routed to massive superconducting electric motors that spin giant fan blades inside the engines (like an electric turbofan). The exhaust is actually cooler than a commercial jet, and with up to 1,000 MW of power, it can take off on a very short runway.
2. **Second Gear (Supersonic):** At Mach 1, the fan blades lock. The engines use microwave emitters to superheat the ramming air (Thermal Ramjet).
3. **Third Gear (Hypersonic):** Past Mach 5, magnetic fields ionize the thin air into a true plasma stream (Plasma Scramjet).
4. **Fourth Gear (Vacuum):** The intakes seal shut, and the engines draw from internal water/hydrogen tanks for the final orbital burn.

An initial guess at vehicle mass is

- **Payload (Crew Dragon):** ~12,000 kg.
- **Spaceplane Chassis & Wings:** ~15,000 kg.
- **Reaction Mass (Water/Hydrogen):** ~20,000 kg.
- **Orbitron Array & Thrusters:** ~13,000 kg.
- **Total Vehicle Mass: 60,000 kg**.

SSTO fusion powered spaceplane concept

Peak power is calculated as follows: As you climb and accelerate, Power = Drag × Velocity. As you approach orbital velocity, centrifugal force cancels out your weight, dropping drag to near zero. **Your peak power demand happens exactly in the middle of this curve.**

- **Peak Power Speed:** ~4,500 meters per second (Mach 13).
- **Apparent Weight:** Centrifugal force reduces your 60-ton weight to an apparent 40 tons.
- **The Drag:** With a Lift-to-Drag ratio of 4:1, you generate roughly **100,000 Newtons** of drag.
- 100,000 Newtons × 4,500 m/s = **450 Megawatts of mechanical power.** Factoring in system inefficiencies, the Orbitron array would need to hit a maximum power output of roughly **800 Megawatts to 1 Gigawatt (GW).**

The first step in getting this done is to imagine it in more detail. The way I have chosen to imagine it is to clone the [Space Shuttle model in FlightGear] and then whittle away at it to get what I want.

Space Shuttle in FlightGear. Click the picture to fly!

What this approach gives us is:

- There is an ISS to fly to
- We get all the calcs needed to model atmospheric flight
- We get all the calcs needed to model transition to orbit and orbital flight
- We have a ton of switches we can customize and whittle down to what we need for this gadget
- These switches remind us of all the other junk we need to implement in a spaceplane. And it's already implemented

We need to make some changes:

- Delete the outboard fuel tank and boosters and move the logic of the boosters into the internal engines
- Reconfigure the tail engines to be 6 SABRE engines
- Model a SABRE engine
- Model an 1mW p-B11 Orbitron fusion reactor and a rack of 1000 of them
- Add an air scoop which can be closed for re-entry during the hot phase
- Pack the reactors, water tank, p-B11 container, writing and SABRE engine guts into the payload bay.
- Customize and edit the switch panels to match our flight program, engines and power source.

- Rethink the whole launch program from a standard runway, but otherwise keep the checklist similar to the original Space Shuttle checklist.

I am working on an Ubuntu 24.04 Linux machine. To start with I installed the following:

- Flight Gear
- Blender
- FlightGear AC3D patch for Blender
- The stock FlightGear SpaceShuttle model

Then (all of the gory details are here):

- I created an empty GitHub repo with the name CatskillsFusionSSTO with basics like LICENSE, README and .gitignore files.
- I cloned this repo locally under a newly created Aircraft folder.
- I copied the SpaceShuttle aircraft files from the FlightGear local files to my repo.
- I renamed the SpaceShuttle files as CatskillsFusionSSTO files and did a bunch of internal file edits to relabel SpaceShuttle as CatskillsFusionSSTO.
- I checked it all back into GitHub

To size the software engineering task, for code we have:

```
EXT      LANGUAGE      LINES
----------------------------------------
.nas     Nasal (Scripting) 92272
.xml     XML (Systems/Logic) 135917
.eff     GLSL Effects    10109
.js      JSBSim (Math)   0
----------------------------------------
TOTAL LOGIC LINES: 238298
```

Code file inventory

For graphics we have:

```
EXT     TYPE        COUNT    SIZE
-------------------------------------------------------
.png    Textures      285      84M
.jpg    Thumbnails    126      6.5M
.ac     3D Models     42       136M
-------------------------------------------------------
TOTAL REPAINT AREA: 91M
```

Graphics inventory

The kinds of files are:

- .nas : This is the "Brain." To make a Fusion SSTO, modify about 5% of this (mostly propulsion and fuel management).
- .xml : This is the "Skeleton." Most of these are simple animations (moving switches) that you won't need to change.
- .ac : There are dozens of these. We only need to touch the ones we want to physically change (like the engine nozzles or the hull).
- .png: This is the "Skin." A single 500MB folder might only have 10-20 "Master" textures that cover the entire exterior. If we find and repaint those, the whole plane transforms.

Here is the task:

1. The FDM (Flight Dynamics Model): gutting the shuttle.xml

The Space Shuttle's FDM is designed adropping the SRBs and External Tank.

- **The Change:** We will need to delete the External Tank and SRB definitions from shuttle.xml.
- **The SABRE Logic:** We will replace the three SSMEs with six engine definitions. Since your SABREs have 4 "gears," we will write **JSBSim Functions** that vary the THRUST based on MACH and ATMOSPHERIC DENSITY.
- **Estimate:** We will modify about **2,000 lines** in the FDM.

2. The Nasal "Brain": The Energy Management System

The Shuttle Nasal code (92k lines) manages **Hydraulics** and **Fuel Cells**.

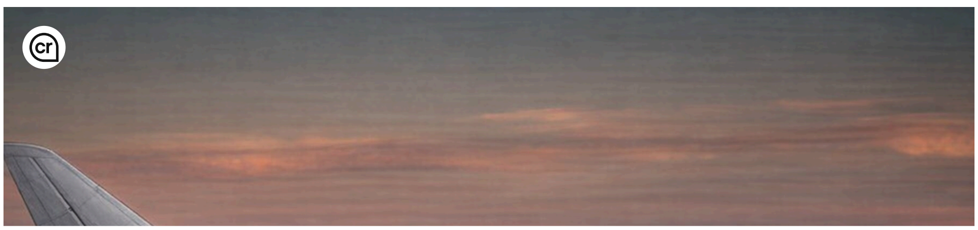
- **New Code:** We need a new script, fusion_core.nas, to manage that **1 GW of power**.
- **The Gears:** We will write a "Mode Controller" that monitors Mach number and toggles between your four gears: **Gear 1 (Electric):**

Divert power to a property like propulsion/engine[n]/fan-rpm. **Gear 2/3 (Thermal/Plasma):** Scale thrust based on the electrical output of the reactor. **Gear 4 (Vacuum):** Switch the JSBSim engine source from Atmospheric Air to Internal Propellant (Water/H2).

- **Estimate:** We will write **3,000–5,000 new lines** of Nasal.

3. Visual Assets: The Blender Surgery

- **3D Models (.ac):** We have 42 models. We only need to touch about **5 or 6**: Delete the SRBs and ET models. Open Models/SpaceShuttle.ac in Blender. Add the 6 SABRE engine nozzles and the large intake scoops. **Payload Bay:** Since the fusion plant is in the bay, "static-mount" a 3D model of the reactor inside Models/PayloadBay/.
- **Textures (.png):** You have 285 files, but 80% are cockpit labels. To get the "Catskills" look, you only need to repaint about **4 or 5 master sheets** (the exterior hull tiles).



Microsoft Copilot

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